# FAST- National University of Computer and Emerging Sciences, Karachi.

## FAST School of Computing, Fall 2025

## CS1005-Discrete Structures

## Assignment # 2

**Instructions:** **Max. Points: 100**

1- This is hand written assignment. You have to submit the hard copy on 31st October 2025 in your class as well scan copy on Google classroom.
2- Just write the question number instead of writing the whole question.
3- You can only use A4 size paper for solving the assignment.
4- You have to write your student ID and Section on the top of each page.

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1. Let R be the following relation defined on the set {a, b, c, d}:

   R = {(a, a), (a, c), (a, d), (b, a), (b, b), (b, c), (b, d), (c, b), (c, c), (d, b), (d, d)}.

   Determine whether R is:

   a) Reflexive  b) Symmetric  c) Antisymmetric
   d) Transitive  e) Irreflexive  f) Asymmetric

2. List the ordered pairs in the relation R from A = {0, 1, 2, 3, 4} to B = {0, 1, 2, 3}, where $(a, b) \in R$ if and only if

   a) $a = b$.  b) $a + b = 4$.  c) $a > b$.
   d) $a \mid b$.  e) G.C.D. (a, b) = 1.  f) L.C.M. (a, b) = 2.

3. List all the ordered pairs in the relation R = {(a, b) | a divides b} on the set {1, 2, 3, 4, 5, 6}.
   Display this relation as Directed Graph (digraph), as well in matrix form.
   Determine whether the relation R is Equivalence or Partial Order Relation.

4. For each of these relations on the set {1, 2, 3, 4}, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

   a) {(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)}  b) {(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)}
   c) {(2, 4), (4, 2)}  d) {(1, 2), (2, 3), (3, 4)}
   e) {(1, 1), (2, 2), (3, 3), (4, 4)}  f) {(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)}

5. Determine whether the relation R on the set of all people is reflexive, symmetric, antisymmetric, Asymmetric, irreflexive and/or transitive, where $(a, b) \in R$ if and only if:

   a) a is taller than b.  b) a and b were born on the same day.
   c) a has the same first name as b.  d) a and b have a common grandparent.

6. Give an example of a relation on a set of Whole Numbers that is:

   a) both symmetric and antisymmetric.  b) neither symmetric nor antisymmetric
   c) both symmetric and transitive  d) neither Symmetric nor transitive

7. Consider these relations on the set of real numbers: A= {1,2,3}

   R1 = $\{(a, b) \in R \mid a > b\}$, the "greater than" relation,
   R2 = $\{(a, b) \in R \mid a \geq b\}$, the "greater than or equal to "relation,
   R3 = $\{(a, b) \in R \mid a < b\}$, the "less than" relation,
   R4 = $\{(a, b) \in R \mid a \leq b\}$, the "less than or equal to "relation,
   R5 = $\{(a, b) \in R \mid a = b\}$, the "equal to" relation,
   R6 = $\{(a, b) \in R \mid a \neq b\}$, the "unequal to" relation.

   Find:

   a) $R2 \cup R4$.  b) $R3 \cup R6$.  c) $R3 \cap R6$.  d) $R4 \cap R6$.  e) $R3 - R6$.
   f) $R6 - R3$.  g) $R2 \oplus R6$.  h) $R3 \oplus R5$.  i) $R2 \circ R1$.  j) $R6 \circ R6$.

8. a) Represent each of these relations on {1 , 2 , 3} with a matrix(with the elements of this set listed in increasing order).

   i) {(1 , 1), ( 1 , 2), ( 1 , 3 }  ii) {(1 , 2), ( 2 , 1), ( 2 , 2) ,( 3 , 3)}
   iii) {(1 , 1), (1 , 2), (1 , 3), (2 , 2), (2 , 3), (3 , 3)}  iv) {( 1 , 3), ( 3 , 1)}

b) List the ordered pairs in the relations on {1 , 2 , 3 } corresponding to these matrices(where rows and columns correspond to the integers listed in increasing order).

i) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ ii) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ iii) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

9. a) Suppose that $R$ is the relation on the set of strings of English letters such that $aRb$ if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string $x$. Is $R$ an equivalence relation?
b) Let $m$ be an integer with $m > 1$. Show that the relation $R = \{(a,b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation on the set of integers.

10. a) Find the first five terms of the sequence for each of the following general terms where n > 0.
i) $2^n - 1$ ii) $10 - \frac{3}{2}n$ iii) $\frac{(-1)^n}{n^2}$ iv) $\frac{3n+4}{2n-1}$
b) Identify the following Sequence as Arithmetic or Geometric Sequence then find the indicated term.
i) -15, -22, -29, -36, ……; 11th term. ii) a - 42b, a - 39b, a - 36b, a - 33b, ….; 15th term.
iii) 4, 3, $\frac{9}{4}$, . . ..; 17th term iv) 32, 16, 8, ….; 9th term

11. a) Find the G.P in which:
i) $T_3 = 10$ and $T_5 = 2\frac{1}{2}$ ii) $T_5 = 8$ and $T_8 = -\frac{64}{27}$
(b) Find the A.P in which:
i) $T_4 = 7$ and $T_{16} = 31$ ii) $T_5 = 86$ and $T_{10} = 146$

12. a) How many numbers are there between 256 and 789 that are divisible by 7? Also find their sum.
b) Find the sum to n terms of an A.P whose first term is $\frac{1}{n}$ and the last term is $\frac{n^2-n+1}{n}$.

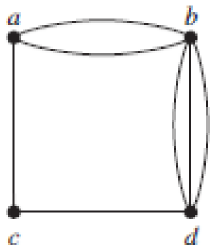
13. a) Use summation notation to express the sum of the first 100 terms of the sequence $\{a_j\}$, where $a_j = \frac{1}{j}$ for $j = 1, 2, 3....$
b) What is the value of:
i) $\sum_{k=4}^{8}(-1)^k$. ii) $\sum_{j=1}^{5}(j)^2$.

14. a) A batsman hits the ball in the air after reaching the maximum height, ball dropped 40 feet during the 1st Second, 33 feet during the 2nd Second, and 26 feet during the 3rd Second, and so on. How many feet the ball dropped in 5th Second. Also find the total distance covered by the ball after 5th second.
b) If the population of a town increases geometrically at the rate of 7% annually and the present population is 70000. What will be the population after 6 years from now?

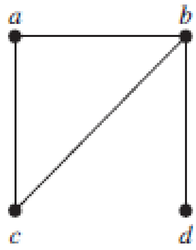
15. a) Find the first six terms of the sequence defined by each of these recurrence relations and initial conditions.
i) $a_n = -2a_{n-1}$, $a_0 = -1$ ii) $a_n = a_{n-1} - a_{n-2}$, $a_0 = 2$, $a_1 = -1$
iii) $a_n = 3a^2_{n-1}$, $a_0 = 1$ iv) $a_n = na_{n-1} + a^2_{n-2}$, $a_0 = -1$, $a_1 = 0$
b) What are the terms $a_0$, $a_1$, $a_2$ and $a_3$ of the sequence $\{a_n\}$, where $a_n$ equals?
i) $(-2)^n$ ii) 3 iii) $7 + 4^n$ iv) $2^n + (-2)^n$

16. Determine whether the graph shown in figure i to iv has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops. Use your answers to determine the type of graph.

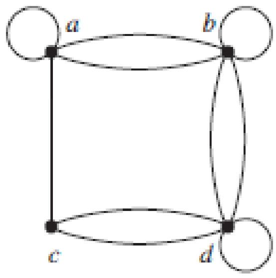
a)



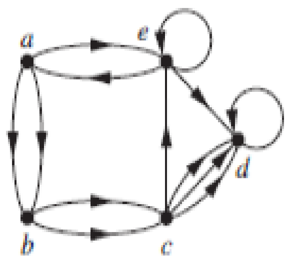

b)




c)



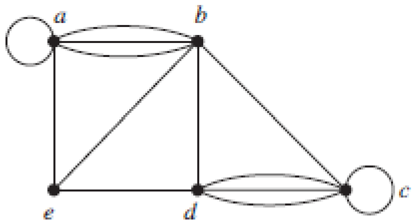

d)



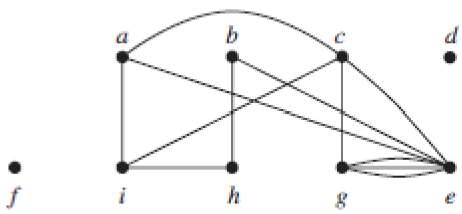

**17.** The intersection graph of a collection of sets A1, A2 ,..., An is the graph that has a vertex for each of these sets and has an edge connecting the vertices representing two sets if these sets have a nonempty intersection. Construct the intersection graph of these collections of sets.
a) A1 = {0, 2, 4, 6, 8}, A2 = {0, 1, 2, 3, 4}, A3 = {1, 3, 5, 7, 9}, A4 = {5, 6, 7, 8, 9}, A5 = {0, 1, 8, 9}
b) A1 = {…,−4,−3,−2,−1, 0}, A2 = {…,−2,−1, 0, 1, 2,…}, A3 = {…,−6,−4,−2, 0, 2, 4, 6,…}, A4 = {…,−5,−3,−1, 1, 3, 5,…}, A5 = {…,−6,−3, 0, 3, 6,…}

**18.** a) Find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Also find the neighborhood vertices of each vertex in given graphs.

i)

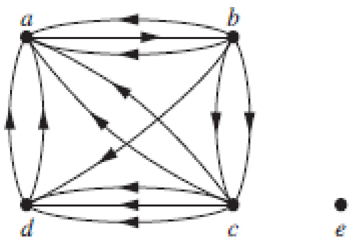



ii)

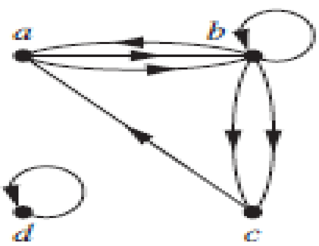



b) Determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph.

i)




ii)

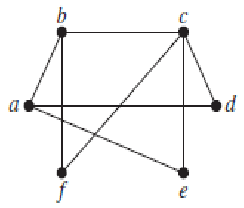



**19.** a) Suppose that a new company has five employees: Zamora, Agraharam, Smith, Chou, and Macintyre. Each employee will assume one of six responsibilities: planning, publicity, sales, marketing, development, and industry relations. Each employee is capable of doing one or more of these jobs: Zamora could do planning, sales, marketing, or industry relations; Agraharam could do planning or development; Smith could do publicity, sales, or industry relations; Chou could do planning, sales, or industry relations; and Macintyre could do planning, publicity, sales, or industry relations. Model the capabilities of these employees using appropriate graph.

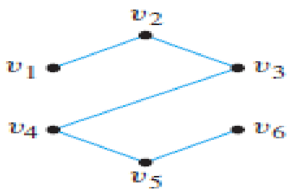
b) Suppose that there are four employees in the computer support group of the School of Engineering of a large university. Each employee will be assigned to support one of four different areas: hardware, software, networking, and wireless. Suppose that Ping is qualified to support hardware, networking, and wireless; Quiggley is qualified to support software and networking; Ruiz is qualified to support networking and wireless, and Sitea is qualified to support hardware and software. Use appropriate graph to model the four employees and their qualifications.

**20.** Find which of the following graphs are bipartite. Redraw the bipartite graphs so that their bipartite nature is evident. Also write the disjoint set of vertices.

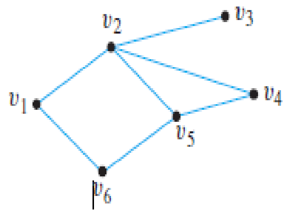
a)



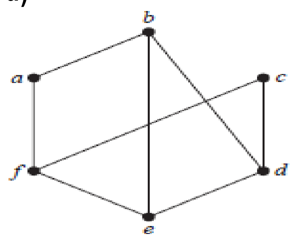

b)



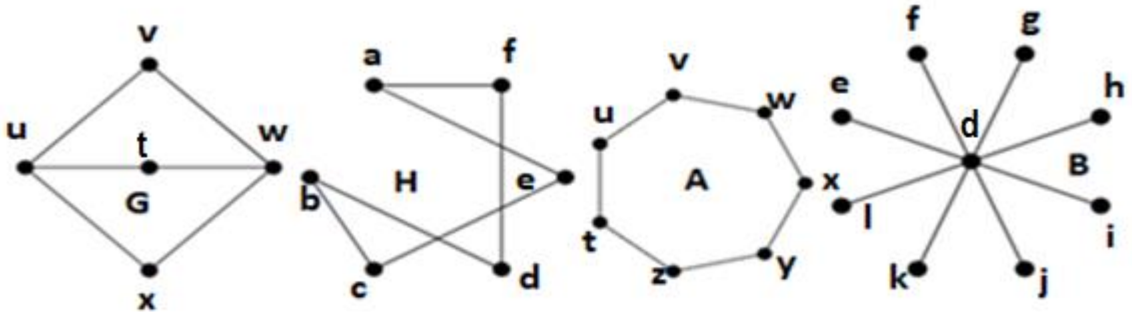

c)




d)

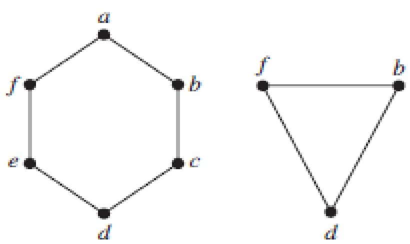



**21.** a) Which of the graphs below are bipartite? Justify your answers.

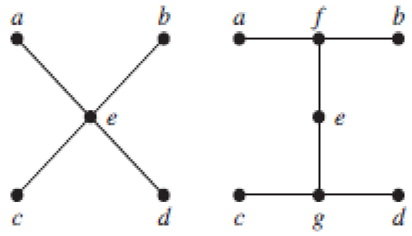



b) Find the union of the given pair of simple graphs. (Assume edges with the same endpoints are the same.)

i)



ii)



**22.** Draw a graph with the specified properties or show that no such graph exists.

a) A graph with four vertices of degrees 1, 1, 2, and 3.

b) A graph with four vertices of degrees 1, 1, 3, and 3.

c) A simple graph with four vertices of degrees 1, 1, 3, and 3.

**23.** a) In a group of 15 people, is it possible for each person to have exactly 3 friends? Explain. (Assume that friendship is a symmetric relationship: If x is a friend of y, then y is a friend of x.)

b) In a group of 4 people, is it possible for each person to have exactly 3 friends? Why?

c) If 10 people each shake hands with each other, how many handshakes took place? What does this question have to do with graph theory?

d) Among a group of 5 people, is it possible for everyone to be friends with exactly 2 of the people in the group? What about 3 of the people in the group?

e) How many vertices does a regular graph of degree four with 10 edges have?

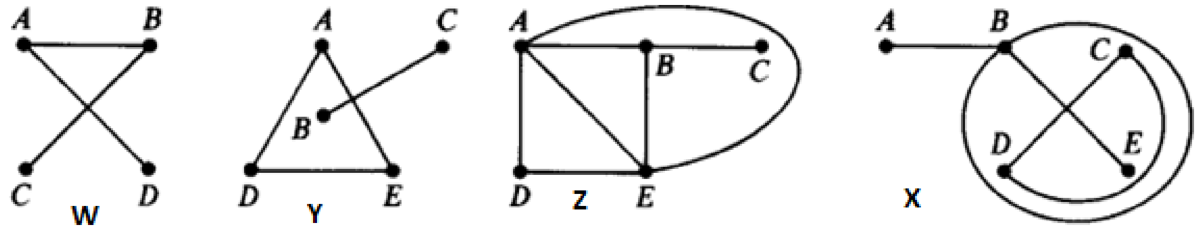
**24.** Consider the multigraphs and find the following.

a) Which of them are connected? If a graph is not connected, find its connected components.

b) Which is/are cycle-free (without cycles)?

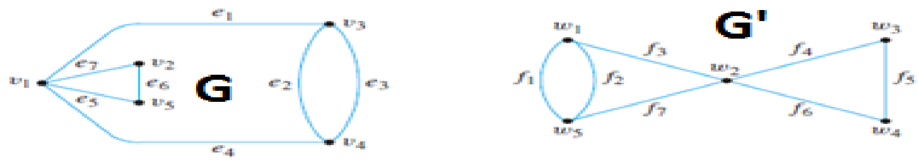
c) Which is/are loop-free (without loops)?

d) Which is/are (simple) graphs?

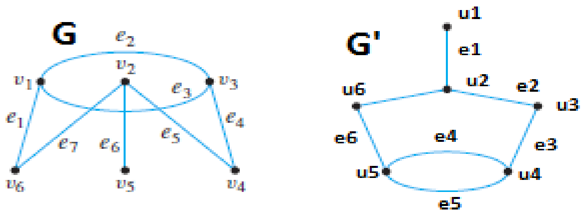



25. For given pair (G, G') of graphs. Determine whether they are isomorphic. If they are, give function g: V (G) →V (G') that define the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share.

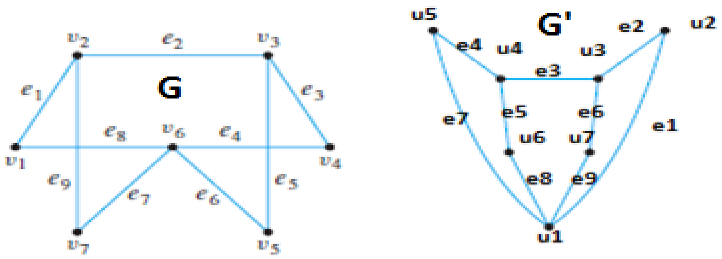
a)



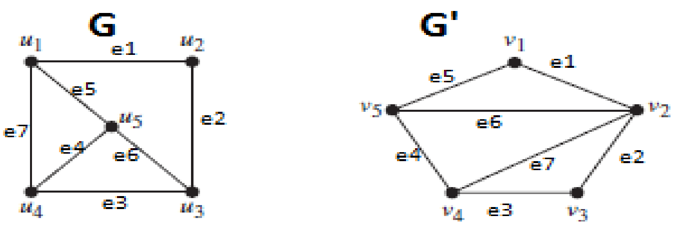

b)



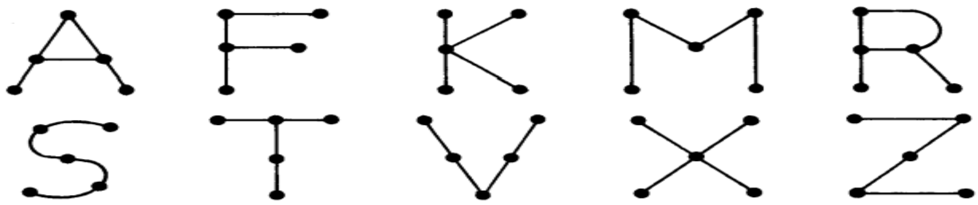

c)




d)

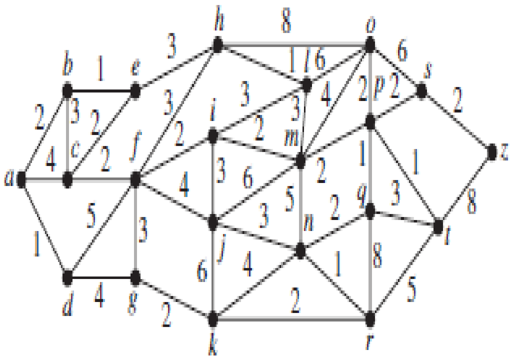



26. Given below are different graphs of Letters, identify which are the isomorphic to each other. Provide the invariant for graph isomorphism that they share or do not share.

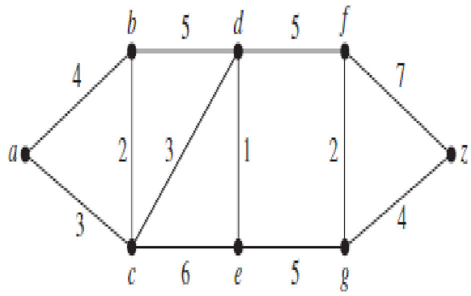


**27.** Find the length of a shortest path between node *a* and node *z* in the given weighted graph by using Dijkstra's algorithm.

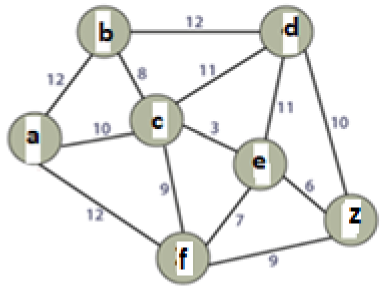
a)



b)

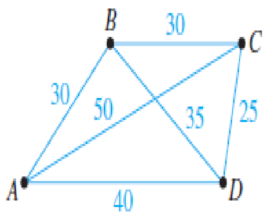



c)

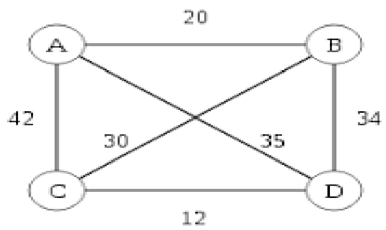



**28.** Imagine that the drawing below is a map showing four cities and the distances in kilometers between them. Suppose that a salesman must travel to each city exactly once, starting and ending in city A. Which route from city to city will minimize the total distance that must be traveled?

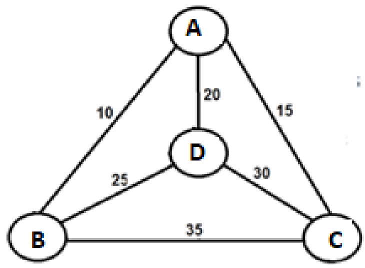
a)



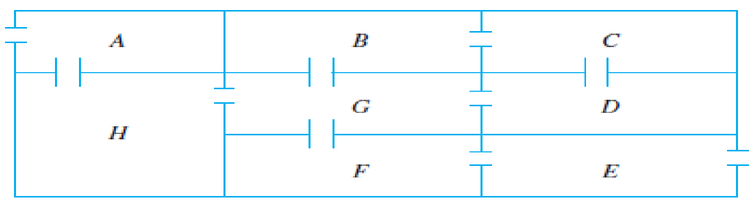

b)



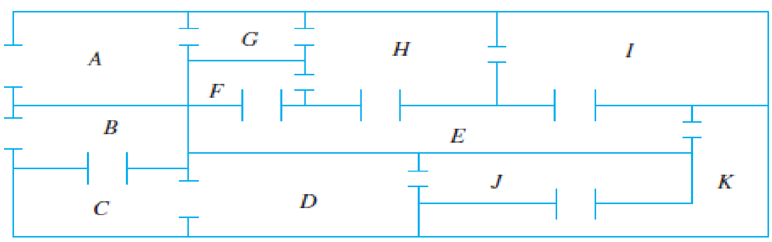

c)




**29.** a) The following is a floor plan of a house. Is it possible to enter the house in room A, travel through every interior doorway of the house exactly once, and exit out of room E? If so, how can this be done?



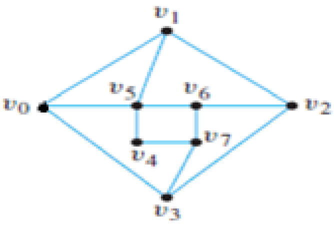

b) The floor plan shown below is for a house that is open for public viewing. Is it possible to find a trail that starts in room A, ends in room B, and passes through every interior doorway of the house exactly once? If so, find such a trail.



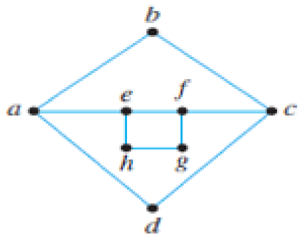

c) You are given a 10 pieces of domino set whose titles have the following set of dots:
(1,2); (1,3); (1,4); (1,5); (2,3); (2,4); (2,5); (3,4); (3,5); (4,5).
Discuss the possibility of arranging the titles in a connect series such that number on a title always touches the same number on it neighbor. (Hint: Use a five vertex complete graph and see if it is an Euler Graph.)

30. a) Find Hamiltonian Circuits and Path for those graphs that have them. Explain why the other graphs do not.

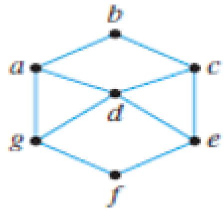
i)




ii)

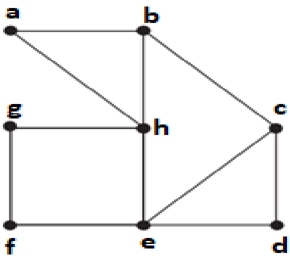



iii)

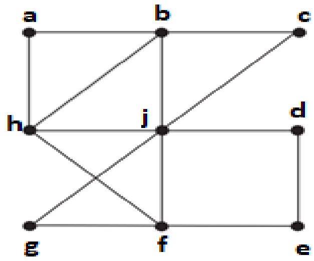



b) Determine which of the graphs have Hamiltonian circuits and Path. If the graph does not have Hamiltonian Circuits and Path, explain why not. If it does have Hamiltonian Circuit and Path, describe one.

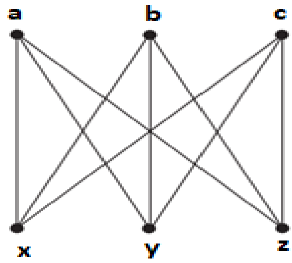
i)




ii)

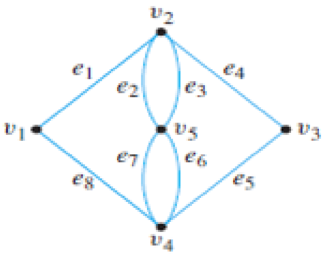



iii)

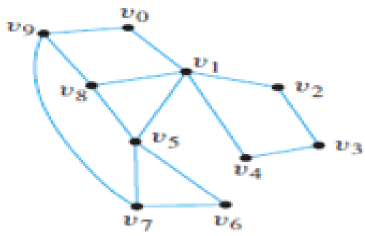



31. a) Determine which of the graphs have Euler circuits. If the graph does not have an Euler circuit, explain why not. If it does have an Euler circuit, describe one.

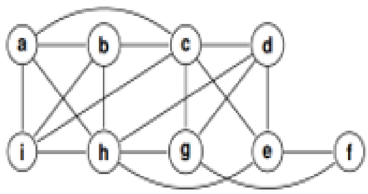
i)




ii)

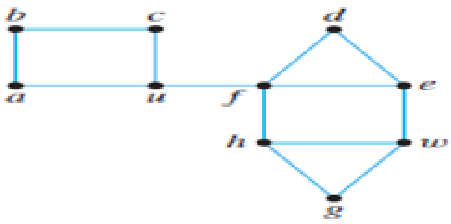



iii)

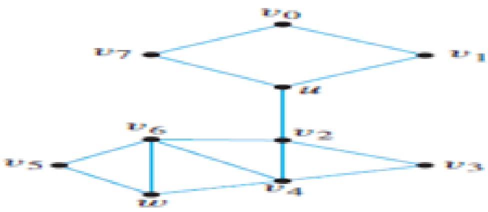



b) Determine whether there is an Euler path from u to w. If the grap h does not have an Euler path, explain why not. If it does have an Euler path, describe one.

i)

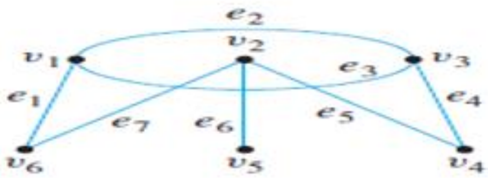



ii)

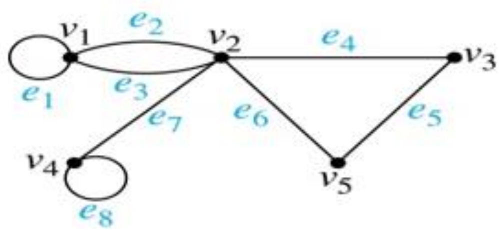



32. a) Use an incidence matrix to represent the graph shown below.

i)




ii)




b) Draw a graph using below given incidence matrix.

i)

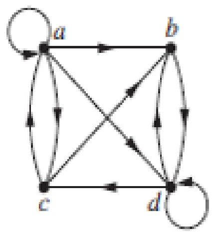
$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

ii)

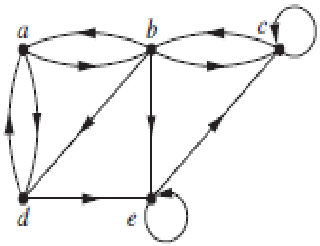
$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

33. a) Use an adjacency list and adjacency matrix to represent the given graph.

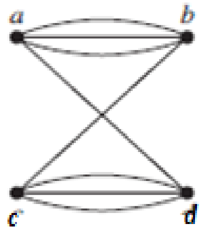
i)



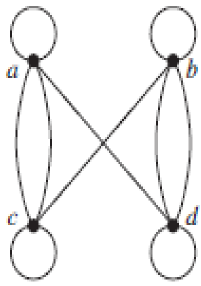

ii)




iii)




iv)

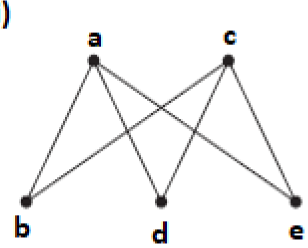



b) Draw the graph represented by the given adjacency matrix.

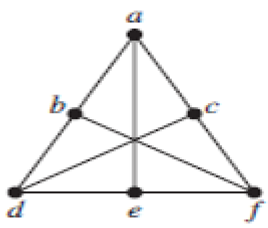
(i) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 2 & 3 & 0 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

34. a) Determine whether the given graph is planar. If so, draw it so that no edges cross.

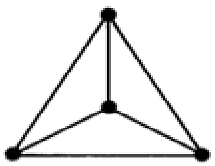
i)



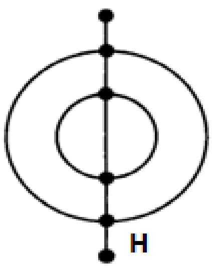

ii)



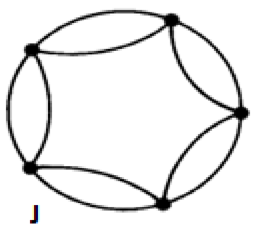

b) Count the number of vertices V, the number of edges E, and the number of regions R of each graph given below and also verify Euler's formula.



G



H

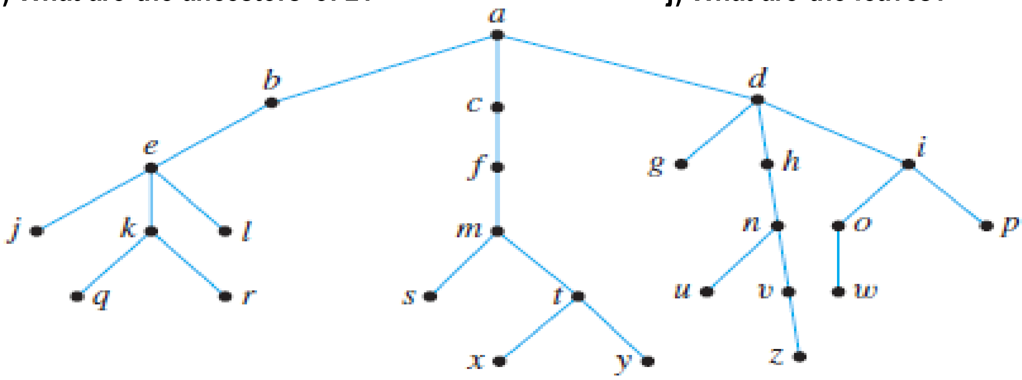


J

35. Consider the tree shown at right with root a.

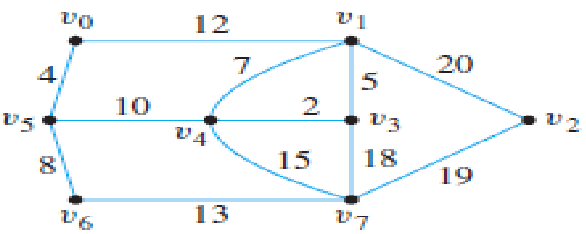
a) What is the level of n?
b) What is the level of a?
c) What is the height of this rooted tree?
d) What are the children of n?
e) What is the parent of g?
f) What are the siblings of j?
g) What are the descendants of f?
h) What are the internal nodes?

i) What are the ancestors of z? j) What are the leaves?

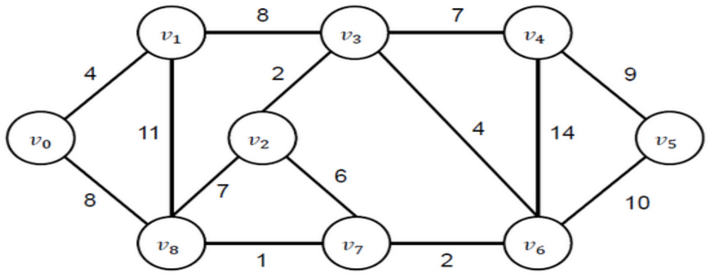



36. a) Use Prim's algorithm to find a minimum spanning tree starting from $V_0$ for given graphs. Indicate the order in which edges are added to form each tree.

i)

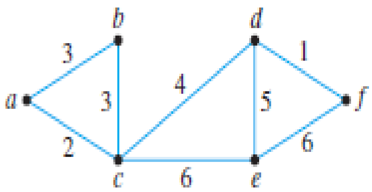



ii)

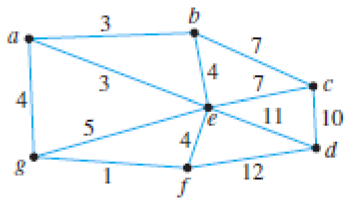



b) Use Kruskal's algorithm to find a minimum spanning tree for given graphs. Indicate the order in which edges are added to form each tree.

i)




ii)

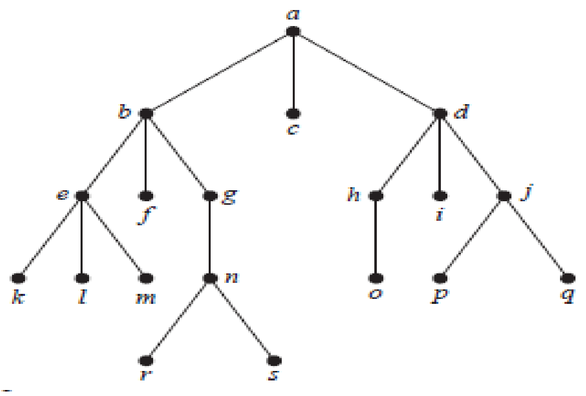



37. (a) Build a binary search tree for the give set of word's using alphabetical order:
   i) banana, peach, apple, pear, coconut, mango, and papaya.
   ii) oenology, phrenology, campanology, ornithology, ichthyology, limnology, alchemy, and astrology.

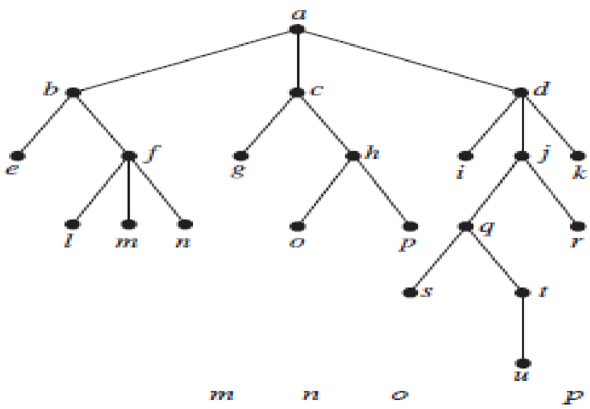
(b) Represent these expressions using binary trees.
   (i) (x + xy) + (x / y)
   (ii) x + ((xy + x) / y)

38. a) Determine the order in which Preorder, Inorder and Postorder traversal visits the vertices of the given ordered rooted tree.

i)

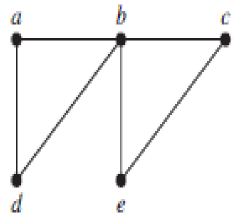



ii)

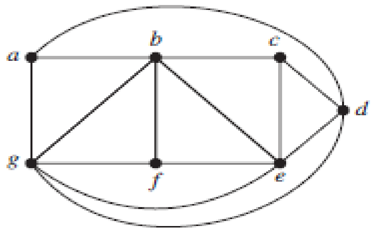



b) Find a spanning tree for the graph shown by removing edges in simple circuits. Write down the removed edges.

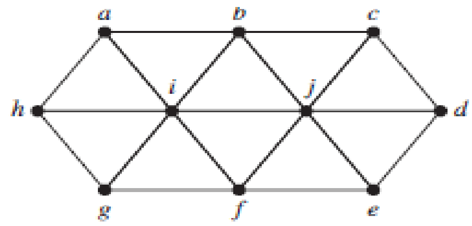
i)




ii)




iii)




39. a) How many edges does a tree with 10000 vertices have?
 b) How many edges does a full binary tree with 1000 internal vertices have?
 c) How many vertices does a full 5-ary tree with 100 internal vertices have?
 d) How many leaves does a full 3-ary tree with 100 vertices have?

40. (a) Write these expressions in Prefix and Postfix notation:
 i) (x + xy) + (x / y)  ii) x + ((xy + x) / y)

 (b)
 i) What is the value of this prefix expression + – ↑ 3 2 ↑ 2 3 / 6 – 4 2
 ii) What is the value of this postfix expression 4 8 + 6 5 – * 3 2 – 2 2 + * /

41. Answer these questions about the rooted tree illustrated.

 a) Is the rooted tree a full m-ary tree?

 b) Is the rooted tree a Balanced m-ary tree?

 c) Draw the subtree of the tree that is rooted at
 i) c.  ii) f.  iii) q.

42. Suppose that a popular company launches "Nothing is like 127.0.0.1" for both men and women. Women T-shirts come in four different sizes: M, L, XL, and XXL. Each size comes in four colors (white, red, green, and black), except XL, which comes only in red, green, and black, and XXL, which comes only in green and black. Men T-shirts come in five different sizes: S, M, L, XL, and XXL. Each size comes in three colors (white, green, and black), except XL and XXL, which comes only in white and black.

 a) Use a tree diagram to determine what is the minimum number of T-shirts that the shop needs to stock to have one of each size and color available for both men and women?

b) Consider the tree obtained in part (a).
Show how many internal vertices and leaves the tree has? Determine the height of the tree?

c) Consider the tree obtained in part (a).
   i) Determine whether it is a Full m-ary tree or not? Give reason.
   ii) Determine whether it is a Balanced m-ary tree or not? Give reason.

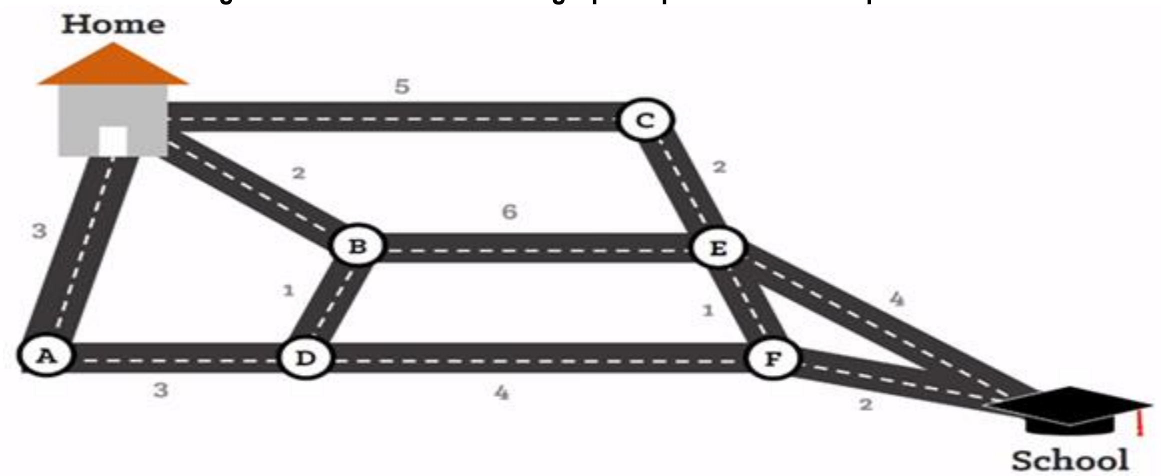
d) Show the Preorder and Post order traversal of the tree obtained in part (a).

43. Consider a scenario from World War II where two Germen soldiers are communicating with each other through a secure channel. Head of Platoon want to communicate two different messages to each of them through separate communication channels. Both of them can received the message but can't reply back. After receiving the message from the head of platoon, these two soldiers communicate the received message to one of their subordinates each. These subordinates can't reply to them but have a one-way communication channel with the Platoon Head so that they can confirm the message delivery to the head of platoon.

a) Transform the given scenario into a digraph. Show the in-degree and out-degree of each vertex.

b) Using diagraph in part a), determine whether the relation is an equivalence relation OR a partial-order relation? Show all of your steps.

44. Musab is an undergraduate student. The below graph map shows different path from his home to the school.




a) Musab wants to determine the shortest path from home to school using Dijkstra Algorithm. What will be the cost of that path? Show all steps.

b) Determine the Euler Circuit and Path in the above graph for Musab? Show all of your steps with reasoning.

c) Determine the Hamilton Circuit and Path in the above graph for Musab? Show all of your steps with reasoning.

45. a) Determine if the following two graphs *G* and *H* as shown in figure# 1 & 2 are isomorphic. If they are, give function *f: V (G) → V (H)* that define the isomorphism. If they are not, give the reason why?

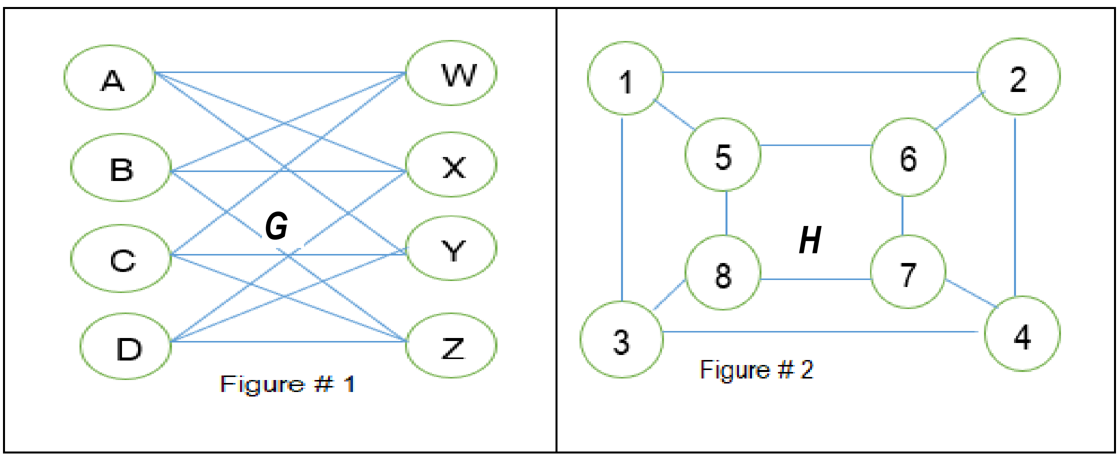



Figure # 1

Figure # 2

b) Consider the graphs *G* and *H* from part (a).

i) Is Graph *G* a complete Bipartite Graph? Explain.

ii) Determine if Euler and Hamilton circuits exist in Graph *H*. If yes, show the circuit, if not explain why?

**46.** Let A= {a, b, c, d} and B = {a, b, c, d}. Consider the following functions:

a) *f(a) = b, f(b) = a, f(c) = c, f(d) = d* b) *f(a) = b, f(b) = b, f(c) = d, f(d) = c*

c) *f(a) = d, f(b) = b, f(c) = c, f(d) = d* d) *f(a) = c, f(b) = a, f(c) =b, f(d) = d*

(i) Determine the Domain, Co-domain and Range of the functions.

(ii) Determine whether the functions are Injective, Surjective and Bijective or not?

(iii) Determine the inverse of function if exists.

**47.** a) *Let* $f(x) = \lfloor \frac{x^2}{3} \rfloor$, Find $f$ (S) if:

(i) S = {-2,-1,0,1,2,3} (ii) S = {0,1,2,3,4,5} (iii) S = {1,5,7,11} (iv) S = {2,6,10,14}

b) Solve the following:

(i) $\lceil \frac{3}{4} \rceil$ (ii) $\lfloor \frac{7}{8} \rfloor$ iii) $\lceil -\frac{3}{4} \rceil$ (iv) $\lfloor -\frac{7}{8} \rfloor$

(v) $\lceil 3 \rceil$ (vi) $\lfloor -1 \rfloor$ (vii) $\lfloor \frac{1}{2} + \lceil \frac{3}{2} \rceil \rfloor$ (viii) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor$

c) Prove or disproof that if $x$ is a real number, then $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.

**48.** Let $f$ and g be the functions from the set of integers to the set of integers defined by $f$(a) = 2a + 3 and g(a) = 3a + 2.

a) What is the composition of $f$ and g? What is the composition of g and $f$?

b) Which type of function $f$ and g are?

c) Are $f$ and g invertible?

**49.** As we have discussed, the practical application of all the topics in the class. Now you are required to submit at least two real world applications of the following topics.

a) Functions
b) Relations
c) Sequence
d) Series
e) Graph theory
f) Trees